

RAVEN ROCK, PA, USA

WMO: 722826

Lat: 39.733N

Lon: 77.433W

Elev: 275

StdP: 98.07

Time Zone: -5.00 (NAE)

Period: 06-17

WBAN: 93762

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.8	(c) -9.4	(d) -20.5	(e) 0.6	(f) -10.1	(g) -18.4	(h) 0.8	(i) -8.0	(j) 7.7	(k) 1.3	(l) 7.1	(m) 0.5	(n) 2.1	(o) 230	(p) 0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	33.4	22.9	31.7	22.1	30.3	21.5	24.6	30.7	23.8	29.3	23.0	28.1	1.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.1	27.5	22.1	17.3	26.8	21.3	16.5	26.3	75.8	30.7	72.5	29.4	69.6	28.3	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.2	5.3	4.6	-14.1	36.3	2.9	3.7	-16.2	39.0	-17.9	41.2	-19.5	43.3	-21.6	46.0		
			-15.1	26.0	2.4	1.2	-16.8	26.8	-18.2	27.5	-19.5	28.2	-21.3	29.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	-0.1	0.9	6.0	12.2	17.4	21.9	24.4	23.5	19.6	13.8	7.9	3.3
	DBStd	9.69	5.53	6.14	5.87	5.41	4.42	3.28	3.16	2.70	3.81	4.53	4.62	5.11
	HDD10.0	1093	318	261	151	36	1	0	0	0	0	16	93	218
	HDD18.3	2685	572	488	382	197	74	8	1	1	31	154	313	466
	CDD10.0	2051	5	6	28	102	230	358	446	418	287	133	29	10
	CDD18.3	603	0	0	1	13	44	115	188	162	67	13	0	0
	CDH23.3	5146	0	2	7	126	371	947	1770	1333	509	79	0	1
	CDH26.7	1731	0	0	0	26	95	284	709	442	159	16	0	0
Wind	WSAvg	1.3	1.8	1.9	1.9	1.8	1.1	1.0	0.9	0.7	0.8	1.2	1.4	1.6
Precipitation	PrecAvg	1118	84	63	105	94	94	101	98	96	116	86	80	89
	PrecMax	1750	212	151	227	192	213	221	245	152	243	192	169	161
	PrecMin	726	31	7	12	43	22	32	44	38	0	12	13	16
	PrecStd	230	45	34	46	38	48	55	47	30	68	52	37	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.3	19.9	24.2	29.7	31.5	33.6	37.2	34.1	33.2	28.8	21.7	19.6
		MCWB	13.5	11.6	14.7	17.1	20.8	23.5	24.7	22.7	22.5	20.1	14.7	15.0
	2%	DB	15.2	16.0	20.9	26.8	29.3	31.3	34.1	32.2	30.7	25.4	19.1	16.5
		MCWB	11.7	8.9	13.1	16.2	20.3	22.1	23.2	22.8	21.5	18.3	13.9	13.5
	5%	DB	11.3	13.1	18.1	24.3	27.3	29.8	32.2	30.6	28.3	23.1	17.1	13.8
		MCWB	8.2	6.9	11.5	15.1	19.0	21.5	22.2	22.2	20.7	17.7	12.2	10.9
	10%	DB	8.0	9.8	15.4	21.6	25.3	28.2	30.4	29.3	26.3	21.1	15.2	10.6
		MCWB	4.4	5.1	10.2	14.1	17.7	20.2	21.6	21.3	19.4	16.0	10.7	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	13.5	16.0	19.7	23.1	24.7	26.0	25.6	24.0	21.9	16.9	16.5
		MCDB	16.9	18.4	21.4	25.0	29.6	31.1	33.7	31.5	29.7	25.2	20.0	18.4
	2%	WB	11.9	9.7	14.7	17.9	21.7	23.6	24.9	24.5	23.0	20.1	14.8	13.7
		MCDB	14.0	14.7	19.0	24.5	27.6	29.7	31.5	29.5	28.3	24.6	17.8	16.1
	5%	WB	8.0	7.6	12.9	16.3	20.6	22.7	24.0	23.5	21.9	18.6	13.1	11.3
		MCDB	11.6	12.9	17.4	22.3	25.6	28.2	29.8	28.0	26.7	21.8	16.6	13.4
	10%	WB	4.5	5.2	10.4	14.9	19.4	21.8	23.1	22.6	20.8	16.9	11.6	7.7
		MCDB	8.0	9.2	15.1	20.5	23.6	26.6	28.4	27.1	24.6	20.0	14.4	10.0
Mean Daily Temperature Range		MDBR	7.4	8.9	9.8	10.4	10.3	10.5	10.4	10.4	9.6	8.6	7.4	
	5% DB	MCDBR	11.3	13.8	13.5	13.9	12.6	11.5	12.2	11.6	12.5	12.2	11.2	10.8
		MCWBR	9.1	9.5	8.4	7.1	6.2	5.6	4.6	4.7	5.5	6.3	7.2	8.6
	5% WB	MCDBR	11.0	12.5	12.1	12.2	11.2	10.4	10.3	9.8	10.6	10.2	9.5	10.7
		MCWBR	9.5	9.3	8.5	7.1	5.9	5.6	4.6	4.7	5.2	6.3	7.2	9.3
Clear-Sky Solar Irradiance	taub	0.304	0.319	0.351	0.393	0.429	0.494	0.512	0.505	0.425	0.361	0.318	0.304	
	taud	2.437	2.376	2.369	2.273	2.202	2.056	2.023	2.022	2.223	2.408	2.493	2.473	
	Ebn at Noon	864	899	903	882	853	796	776	769	813	839	842	837	
	Edh at Noon	81	99	111	130	143	166	170	166	127	95	76	73	
All-Sky Solar Radiation	RadAvg	1.88	2.77	3.68	4.75	5.31	5.89	5.85	5.19	4.23	3.07	2.18	1.63	
	RadStd	0.16	0.33	0.23	0.40	0.58	0.47	0.34	0.27	0.55	0.35	0.20	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (64 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+67

Nomenclature: See separate page